

Does synthetic EHR training fail under external multicenter validation?

Yes, **synthetic EHR training** often **fails to generalize** under external multicenter validation, though not universally.

Does synthetic EHR training fail under external multicenter validation? N = 10

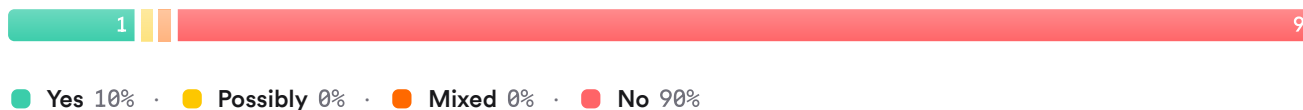


FIGURE 1 Consensus on synthetic EHR external multicenter generalization

The meter should be read as **moderate yes**, not absolute yes. Synthetic-data-trained models usually hold up on same-source or closely matched validation, but external performance degrades once hospitals differ in feature distributions, data completeness, and clinical workflows (Achterberg et al., 2024; Chen et al., 2024; ■ Azam & Singh, 2025).

Synthetic EHR papers directly support **within-source utility**, while real cross-institution studies show why external transport is hard. The exact MIMIC-to-eICU synthetic failure pattern is only sparsely documented, but the combined evidence points to domain shift, sparsity, and protocol variation as the main failure modes (■ Yoon et al., 2023; ■ Azam & Singh, 2025; Tsai et al., 2025).

Synthetic EHR Utility

Evidence

Strength

Claim



Moderate

Models trained on **synthetic data from one source** can perform similarly to real-data-trained models when tested on matched real data from that same source, including MIMIC-based mortality and trajectory tasks (Achterberg et al., 2024; Nikolentzos et al., 2022)



Moderate

Synthetic generation studies usually validate on **same-center or closely related cohorts**, not true multicenter external deployment, so apparent success does not establish transportability (■ Baowaly et al., 2018; Achterberg et al., 2024; Yan et al., 2022)



Moderate

Reviews conclude that synthetic EHR evaluation still lacks **clinical validation and standardized benchmarks**, which limits confidence in external multicenter claims (Budu et al., 2024; Yan et al., 2022; Chen et al., 2024)

FIGURE 2 Evidence strength for synthetic EHR utility claims

External Validation Failures

The clearest direct multicenter failure comes from a 2025 cross-institution blood pressure study: internal MIMIC performance fell markedly on eICU, and **hypotension detection sensitivity dropped to 0%** under external validation (■ Azam & Singh, 2025). That study traced the failure to site-specific feature distributions, population differences, and institutional measurement protocols, and simple recalibration or covariate-shift correction did not solve it (■ Azam & Singh, 2025).

That pattern is consistent with broader EHR evidence showing that internal validation does not ensure broader applicability and that external validation is essential because data quality, standardization, and geography vary across systems (Tsai et al., 2025). Temporal-shift studies also show that out-of-distribution performance decays unless models learn more global representations, with average AUROC decay of **7%** for a baseline count model versus **3%** for a transformer foundation model over 5-9 years (■ Guo et al., 2022).

Why Models Break

- **Measurement sparsity** limits synthetic fidelity for rare features, especially when many MIMIC variables are observed less than once per visit (■ Yoon et al., 2023).
- GAN training is **unstable**, with mode collapse and mode drop that can miss clinically important subpopulations and correlations (Yan et al., 2023; Yan et al., 2022).
- Most EHR foundation and synthetic studies rely on **narrow datasets** such as MIMIC, which weakens claims of broad deployment readiness (Wornow et al., 2023).
- External robustness improves when models are trained or adapted on **multicenter data**, not just a single center or its synthetic proxy (■ Zhang et al., 2026; ■ Guo et al., 2023).

Exceptions and Limits

Not every external test fails. A single-center heart-failure model retained an external C-statistic of **0.79** on a large national synthetic VA cohort, down from 0.91 internally, suggesting partial transport rather than collapse (■ McGilvray et al., 2025). Cross-database ICU transfer can also be modest in some tasks, with MIMIC-IV versus eICU AUC changes as small as +0.019 or -0.015 using a harmonized extraction pipeline (Liao & Voldman, 2023).

Some newer synthetic benchmarks even report that GAN-based methods perform competitively on MIMIC-III and similarly on MIMIC-IV for transportability-oriented evaluations (Chen et al., 2024). But those are still **MIMIC-family evaluations**, not strong evidence that single-center synthetic training survives genuinely heterogeneous multicenter deployment.

So the answer is **yes in general**: synthetic EHR training often fails under external multicenter validation when the synthetic data inherit single-center distributions, but the evidence is not universal and the strongest direct demonstrations still come from real-data cross-site validation studies rather than synthetic-only trials.

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